

Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738213653536-Chapter 7 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.



UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

IS THERE A BETTER WAY FOR JENNY?

Have you ever heard a siren telling you to evacuate? I'm happy it was a malfunction, but I'm worried about what will happen to Aunt Jenny, who is hearing impaired if something like this happens. Isn't there a better way for Jenny?





HELPING DAD START HIS DAY

My dad is hearing impaired.

He has to go to work early in the morning, but he always has trouble getting up.

So, I hear the alarm and wake my dad up.

I hope my dad can go to work without me.



- **How can we express fire conditions when creating an evacuation system when a fire breaks out?**

An environment module measures the temperature, but since measuring the actual temperature is too dangerous, try using a dial module that can replace the set temperature.



TIPS

Let's code it if only one of the two modules' conditions is matched.

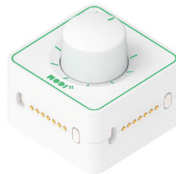


Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



BUTTON



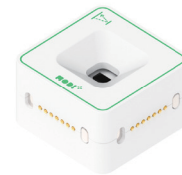
DIAL



ENVIRONMENT



JOYSTICK



TOF



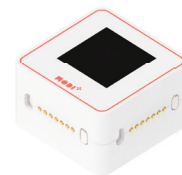
SPEAKER



IMU



LED



DISPLAY



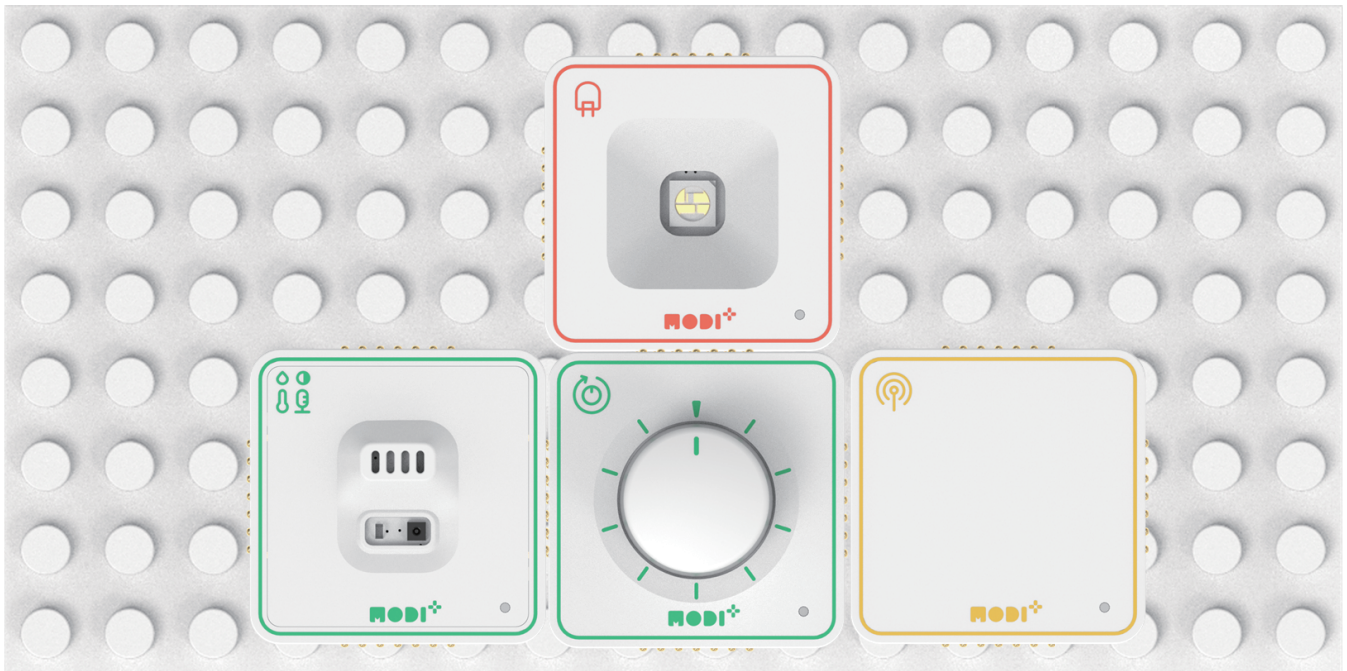
MOTOR A



MOTOR B



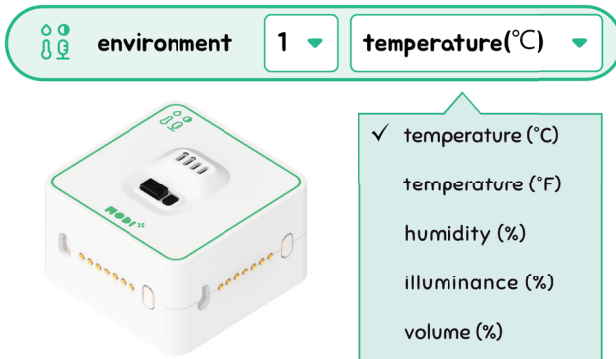
HOW DOES THE CREATION LOOK LIKE?



Let's create a fire visual alarm using the environmental and dial modules' temperature sensors for testing.

Module Functionality

Let's check the details of the input module below for this project.



ENVIRONMENT

- Temperature: Measures how hot or cold the surroundings are.
- Humidity: Checks the amount of moisture in the air.
- Illuminance: Gauges the brightness of the light around it.
- Volume: Detects the level of sound in the environment.

The Environment Module is a smart tool for farmers to make sure their plants are healthy and happy. It checks on three key things plants need:

1. TEMPERATURE MEASUREMENT

Just like we feel too hot or too cold, plants do too! This module keeps an eye on how warm or cool it is, so plants are always comfy.

2. HUMIDITY MEASUREMENT

Plants need just the right amount of water in the air around them. This part of the module measures the air's wetness to help keep plants from getting too thirsty or too soggy.

3. LIGHT MEASUREMENT

Plants need light to grow, and this gadget checks if they're getting enough sunshine, making sure they have the energy to grow big and strong.

4. NOISE MEASUREMENT

While plants don't have ears, they can be affected by the sounds around them. This feature checks how loud or quiet it is, ensuring the plants have a peaceful environment to grow in.



Nested If Statement

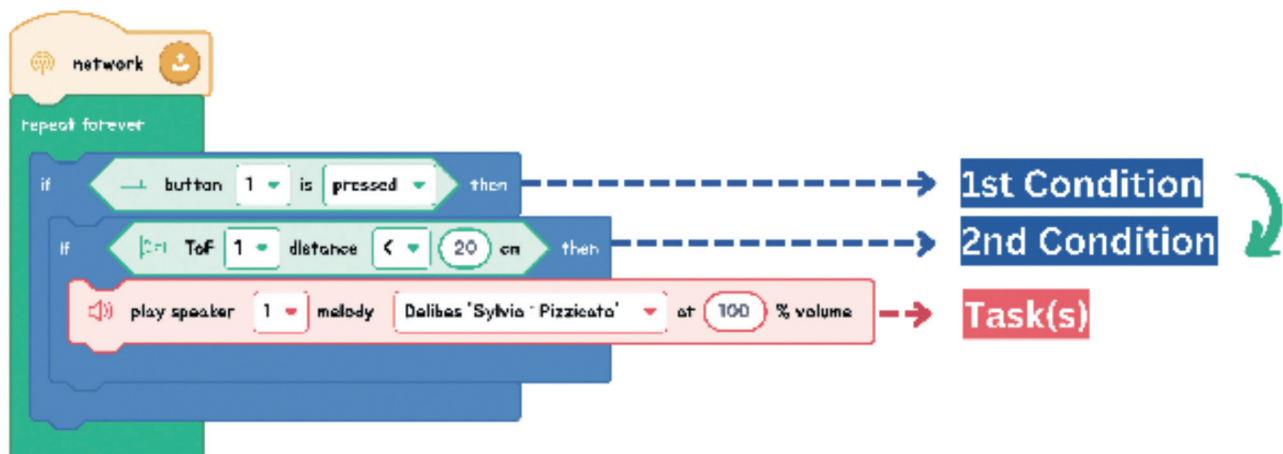
Nested if statements are like putting one decision inside another. First, you check a condition; if that's true, you check another one inside it. If this second condition is also true, then an action takes place. It's like solving a layered puzzle: solve one layer to get to the next!

✓ NESTED IF



First, you check if you're hungry (1st condition). If you are, you then look inside the fridge to see if there's food (2nd condition). If there's food available, you go ahead and make dinner (this is the task).

So, with a 'Nested If', you first decide if you're hungry, and only then do you consider whether there's food in the fridge to make a meal. Let's look at the example below!

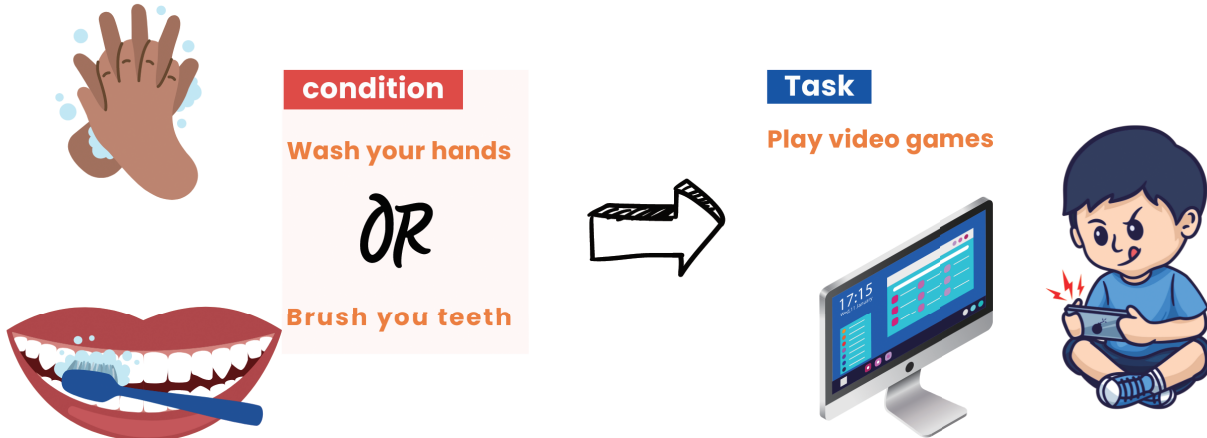


First, the program checks if a button is pressed (1st condition). If the button is indeed pressed, it then examines another condition: whether the distance measured by the ToF sensor is less than 20 cm (2nd condition). Only if both these conditions are met, does the program command the speaker to play the song.

OR

"Or" is like having choices. If you're deciding on a snack and say, "I can have an apple OR a banana," it means you're okay with either one, and you might even choose both if you're extra hungry!

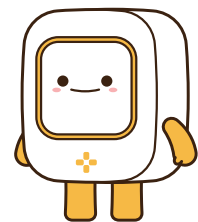
✓ OR



In programming, "and" means both things need to be true, but "or" gives you options—only one thing needs to be true. This helps programmers decide what their code should do in different situations.

TIPS

With "or," it's key to think about how it makes decisions in your code. If any part of an "or" check is true, the whole check is considered true.





Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART

